



Fluids and Lubricants Specifications

Gas engines and gas engine-generator sets

Series 4000 gas engine – marine application

Series 4000 gas engine – generator application and engine-generator set

Series 400 gas engine – engine-generator set

Series 500 gas engine – engine-generator set

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A Rolls-Royce
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Table of Contents

1	Preface			
1.1	General information	5		
2	Lubricants			
2.1	General Information	7		
2.1.1	Engine oil	7		
2.1.2	Engine oil limit values for Series 4000 and Series 400	8		
2.1.3	Engine oil limit values Series 500	10		
2.1.4	Silicon compounds in the fuel gas	11		
2.1.5	Fluorescent dyestuffs for detecting leaks in the lube oil circuit	12		
2.1.6	Lubricating greases	13		
2.1.7	Lubricating greases for general applications	14		
2.2	Series 4000 gas engine – Generator application and engine-generator set - marine application	15		
2.2.1	Approved engine oils	15		
2.2.2	Lubricating greases for generators	18		
2.2.3	Gear oils	19		
2.3	Series 400 gas engine – Engine-generator set	20		
2.3.1	Approved engine oils	20		
2.3.2	Engine oil change intervals	23		
2.3.3	Lubricating greases for generators	25		
2.4	Series 500 Gas Engine - Engine-Generator Set	26		
2.4.1	Approved engine oils	26		
2.4.2	Engine oil change intervals	27		
2.4.3	Oil volume	28		
2.4.4	Lubricating greases for generators	29		
3	Coolants			
3.1	General Information	30		
3.1.1	Coolant – Definition	30		
3.1.2	Operational monitoring / coolant conditioning for Series 4000 and 400 engines	32		
3.1.3	Coolant concentrates – Storage capability	36		
3.1.4	Color additives to detect leakage in the coolant circuit	37		
3.1.5	Preventing damage to cooling system	38		
3.1.6	Unsuitable materials in the coolant circuit	39		
3.1.7	Fresh water requirements – Series 4000	40		
3.1.8	Fresh water requirements – Series 400	41		
3.1.9	Fresh water requirements Series 500	42		
3.2	Series 4000 gas engine – Marine application	43		
3.2.1	Coolant – General information	43		
3.2.2	Coolants without antifreeze – Concentrates for cooling systems free of light metal	44		
3.2.3	Coolant without antifreeze – Ready mixtures for cooling systems free of light metal	46		
3.2.4	Antifreeze – Concentrates for cooling systems free of light metal	47		
3.2.5	Antifreeze – Concentrates for special applications	50		
3.2.6	Antifreeze – Ready mixtures for cooling systems free of light metals	51		
3.3	Series 4000 gas engine – Generator application and engine-generator set	53		
3.3.1	Coolant – General information	53		
3.3.2	Coolant without antifreeze – Concentrates for cooling systems containing light metal	54		
3.3.3	Coolant without antifreeze – Ready mixtures for cooling systems containing light metal	55		
3.3.4	Antifreeze – Concentrates for cooling systems containing light metal	56		
3.3.5	Antifreeze – Concentrates for special applications	60		
3.3.6	Antifreeze – Ready mixtures for cooling systems containing light metals	61		
3.4	Series 400 gas engine – Engine-generator set	64		
3.4.1	Approved coolants	64		
3.5	Series 500 Gas Engine - Engine-Generator Set	65		
3.5.1	Coolants – General information	65		
3.5.2	Coolant – Check and replacement	66		
3.5.3	Approved coolants	67		
4	Fuels			
4.1	General Information	68		
4.1.1	Fuels – Usage	68		
4.1.2	Requirements for fuel gas	69		
4.1.3	Main constituents of natural gas and fuel gases of biogenic origin	70		
4.1.4	Liquid Natural Gas (LNG)	71		
4.1.5	Silicon and sulfur compounds in the fuel gas	72		
4.2	Series 4000 gas engine – Marine application	73		
4.2.1	General	73		
4.2.2	Requirements of fuel gas for marine applications	74		
4.2.3	Specifications for the medium in the gas piping leak chamber for marine applications	77		
4.3	Series 4000 gas engine – Generator application and engine-generator set	79		
4.3.1	General	79		
4.3.2	Natural gas – Requirements of the fuel gas	80		

4.3.3 Biogas – Requirements for fuel gas	83	6.2.1 General information	108
4.4 Series 400 gas engine – Engine-generator set	87	6.2.2 Approved cleaning agents	109
4.4.1 Natural gas – Fuel values	87	6.2.3 Engine cooling system – Cleaning	110
4.4.2 Biogas – Fuel values	88	7 Inlet Air and Combustion Air	
4.4.3 Harmful impurities (pollutants)	89	7.1 General information	111
4.5 Series 500 Gas Engine - Engine-Generator Set	91	7.2 Humidity	112
4.5.1 Gas types	91	7.3 Dust load	113
4.5.2 Minimum requirements of gas composition	92	8 Heating Water	
4.5.3 Gas operating data	93	8.1 General information	114
4.5.4 Conversion of limit values from gas analysis	94	8.2 Information on heating water for Series 4000 and 400	115
4.5.5 Special limit values	95	8.3 Information on Heating Water for Series 500 Engines	117
4.5.6 Operational malfunctions due to condensate in the fuel gas	97	8.3.1 Water quality for heating water circuit Series 500	117
5 Exhaust Gas Aftertreatment System		8.3.2 Heating water – Water quality Series 500	118
5.1 General	98	9 Confirmation for Fluids and Lubricants	
5.2 NOx reducing agent AUS 32 for SCR exhaust gas aftertreatment systems	99	9.1 Confirmation by the operator of engine-generator sets	119
5.3 Exhaust condensate	101	10 Revision Overview	
6 Flushing and Cleaning Specifications for Engine Coolant Circuits		10.1 Revision overview	120
6.1 Flushing and Cleaning Specifications for Series 400 and 4000 Engines	102	11 Appendix A	
6.1.1 General information	102	11.1 List of abbreviations	121
6.1.2 Approved cleaning agents	103	11.2 Conversion table of SI units	122
6.1.3 Engine coolant circuits – Flushing	104	11.3 Contact person/Service partner	123
6.1.4 Engine coolant circuits – Cleaning	105	12 Appendix B	
6.1.5 Cleaning engine coolant circuit assemblies	106	12.1 Index	124
6.1.6 Coolant circuits contaminated with bacteria, fungi or yeast	107		
6.2 Flushing and Cleaning Specifications for Series 500 Engines	108		

1 Preface

1.1 General information

These Fluids and Lubricants Specifications contain general instructions for the proper and safe operation of your product from the manufacturer Rolls-Royce Solutions.

Used symbols and means of representation

The following instructions are highlighted in the text and must be observed:

Important

This field contains product information which is important or useful for the user. It refers to instructions, work and activities that have to be observed to prevent damage or destruction to the material.

Note:

A note provides special instructions that must be observed when performing a task.

Fluids and lubricants

The service life, operational reliability and function of the drive systems are largely dependent on the fluids and lubricants employed. The correct selection and treatment of these fluids and lubricants are therefore extremely important. This publication specifies which fluids and lubricants are to be used.

mtu ValueCare Portfolio

With mtu ValueCare, Rolls-Royce Solutions offers approved oils and coolant tuned to the engine.

Test standard	Designation
DIN	Federal German Standards Institute
EN	European Standards
ISO	International Standards Organization
ASTM	American Society for Testing and Materials
IP	Institute of Petroleum
DVGW	German Gas and Water Industry Association

Table 1: Test standards for fluids and lubricants

Important

Approved fluids and lubricants may not be mixed.

Important

The customer must comply with the instructions specified in the safety data sheets of the respective manufacturers.

Applicability of this publication

The Fluids and Lubricants Specifications will be amended or supplemented as necessary. Prior to use, ensure that the most recent version is available. The most recent version can be consulted under:

<http://www.mtu-solutions.com>

If you have any questions, your contact person will be happy to help you.

Warranty

Use of the approved fluids and lubricants, either under the brand name or in accordance with the specifications given in this publication, constitutes part of the warranty conditions.

The supplier of the fluids and lubricants is responsible for the worldwide standard quality of the named products.

Important

Fluids and lubricants for drive systems may be hazardous materials. Certain regulations must be obeyed when handling, storing and disposing of these substances.

These regulations are contained in the manufacturers' instructions, legal requirements and technical guidelines valid in the individual countries. Great differences can apply from country to country and a generally valid guide to applicable regulations for fluids and lubricants is therefore not possible within this publication.

Users of the products named in these specifications are therefore obliged to inform themselves of the locally valid regulations. Rolls-Royce Solutions accepts no liability whatsoever for improper or illegal use of the fluids and lubricants which it has approved.

When handling fluids and lubricants the "Rules for the protection of the environment" (see Operating Instructions, Safety Regulations, Disassembly and Disposal) must be observed since they are hazardous to health and flammable.

Incorrect use of fluids and lubricants causes environmental pollution:

- Fluids and lubricants must not enter the ground or the sewerage system.
- Used fluids and lubricants must be disposed of through used oil recycling or hazardous waste disposal.
- Used filter elements and cartridges must be disposed of with hazardous waste.

Important

The customer / operator bears the responsibility for observing the fuel values.

Preservation

All information on preservation, represervation and depreservation including the approved preservatives is available in the Preservation and Represervation Specifications (publication number A001070/...). The most recent version can be consulted under:

<http://www.mtu-solutions.com>

2 Lubricants

2.1 General Information

2.1.1 Engine oil

The selection of a suitable engine oil for gas engines depends primarily on the type of gas used to power the engine. The gas engine must only be operated with approved engine oil.

Approved engine oils are specified in the following chapters:

- For Series 4000 (→ Page 15)
- For Series 400 (→ Page 20)
- For Series 500 (→ Page 26)

An essential factor is the share of interfering contamination in the fuel gas. This requires that the operator regularly carries out gas checks. The gas engine oils to be used are characterized by the lowest possible ash content. This prevents increased ash deposits which can lead to reduced catalytic converter performance or combustion knocking.

During operation with biogas, in certain circumstances the engine oil contains corrosive contaminants which are created when the pollutants contained in the gas (chlorine, fluorine and sulfur compounds) are burned. These corrosive constituents can only be neutralized to a limited extent even by special additives engine oil.

Corrosion damage to the oil-lubricated engine components can only be avoided by more frequent oil changes. To improve buffering of concentration peaks when oil is subjected to corrosive contaminants, an increased engine oil volume is urgently recommended.

Important

Dispose of used fluids and lubricants in accordance with local regulations.

Important

Viscosity grade SAE 40 is stipulated for gas engines.
Multigrade oils are not permitted.

Important

Mixing different engine oils is strictly prohibited.

Important

Changing to another oil grade can be accomplished in the course of an oil change provided certain conditions are met. Contact the Service of Rolls-Royce Solutions.

Important

For applications with biogas, sewage gas or landfill gas, the quantity of oil in the engine oil pan is not adequate. A larger volume of oil is required.

2.1.2 Engine oil limit values for Series 4000 and Series 400

Oil change intervals for gas engines

Engine oil change intervals depend on the engine-oil quality, its conditioning, the operating conditions and the fuel used.

An oil sample must therefore be extracted on a regular basis depending on the engine oil volume, type of gas and series and the oil analysis must be compared with the limit values from (→ Table 2). The oil samples must always be taken under the same site conditions (engine at operation temperature) and at the designated point (extraction nozzle on oil filter housing).

If the limit values in accordance with (→ Table 2) are reached or exceeded, an oil change must be carried out immediately.

When using an extended volume of oil, the limit values for wear elements must be reduced inversely proportional to the volume enlargement. The maximum permissible reduction of limit values for the wear elements is 50% of the limit value (→ Table 2).

Fixed oil change intervals without oil analysis are permitted following consultation with Rolls-Royce Solutions.

Limit values for used gas engine oils SAE 40

	Test method	Limit values for Series 4000	Limit values for Series 400
Viscosity at 100 °C (mm ² /s)	ASTM D445 DIN 51562	Max. 17.5 Min. 11.5	New oil value +30 % **
Total base number TBN (mgKOH/g)	ASTM D2896 ISO 3771	Min. 2.5 and TBN > TAN	New oil value -60 % **
Total acid number, TAN (mgKOH/g)	ASTM D664	max. new oil value +2.5	On request from factory
iph value	ASTM D7946	Min. 4	Min. 4
Water (% by vol.)	ASTM D6304 EN 12937 ISO 6296	Max. 0.2	Max. 0.2
Glycol (mg/kg)	ASTM D2982	Max. 100	On request from factory
Oxidation (A/cm)	DIN 51453	Max. 20	Max. 30
Nitration (A/cm)	DIN 51453	Max. 20	Max. 30
Wear elements (mg/kg):	DIN 51399-1/-2		
Iron (Fe)		Max. 30	Max. 50
Lead (Pb)		Max. 20	Max. 30
Aluminum (Al)		Max. 10	Max. 20
Copper (Cu)		Max. 20 ***	Max. 50
Tin (Sn)		Max. 5	Max. 15
Silicon (Si)		Max. 15 *	Max. 10 *

* The limit value for the wear element Si only refers to natural gas operation.

** New oil value can be requested from the factory

*** During the 1st oil change (max. 3000 op. hrs.), the limit value is 50 ppm.

Table 2: Limit values for used gas engine oils SAE 40

Used-oil analysis

The results of the oil analyses must be archived.

From the indicated test methods and limit values, (→ Table 2) you can derive when the result of a single oil sample analysis is to be viewed as abnormal. An abnormal result (e.g. increased wear of the oil) requires an immediate investigation and rectification of the irregular operating condition (e.g. check of the gas preparation or analysis of the gas samples).

The limit values relate to individual oil samples. If the limit values are reached or exceeded, an oil change must be carried out immediately. The results of the oil analysis do not necessarily indicate the wear taking place on specific elements and components.

Aside from the analytical limit values, the condition, operating condition and possible malfunctions of the engine and the periphery of the system are also of equal importance.

2.1.3 Engine oil limit values Series 500

Limit values for used gas engine oils

The following limit values must be observed when determining the oil change intervals for the corresponding series and modules.

Important		
With an installed lube oil volume extension, the following limit values for iron, lead, aluminum, tin, chromium, copper and silicon must be halved in each case. The reason for the limit value adaptation is the greater mixing of particles in the total oil filling volume. Without an adaptation of the limit values the measures for damage prevention can not be introduced.		
Characteristics	Test method	Limit values Series 500
Viscosity at 100 °C (mm ² /s)	ASTM D445 DIN 51562	Max. 17.5
Total base number TBN (mgKOH/ g)	ASTM D2896 ISO 3771	Min. 2.5
Total acid number TAN (mgKOH/ g)	ASTM D664	Fresh oil valve +2.5
iph value	ASTM D7946	Min. 4
Water (% by vol.)	ASTM D6304 EN 12937 ISO 6296	Max. 0.2
Glycol (vol. %)	ASTM D2982 DIN 51375	Max. 0.02
Oxidation (A/cm)	DIN 51453	Max. 20
Nitration (A/cm)	DIN 51453	Max. 20
Wear elements	DIN 51399-1/-2 DIN 51396	
Iron (Fe), ppm per 1000h		Max. 20
Lead (Pb), ppm per 1000h		Max. 35
Aluminum (Al), ppm per 1000h		Max. 15
Copper (Cu), ppm per 1000h		Max. 15
Tin (Sn), ppm per 1000h		Max. 10
Silicon (Si), ppm		Max. 15*
Chromium (Cr), ppm per 1000h		Max. 5
Sodium (Na), ppm		Max. 20
* For sewage gas applications, the limit value is increased to 100 mg/kg		

Table 3: Limit values for used gas engine oils